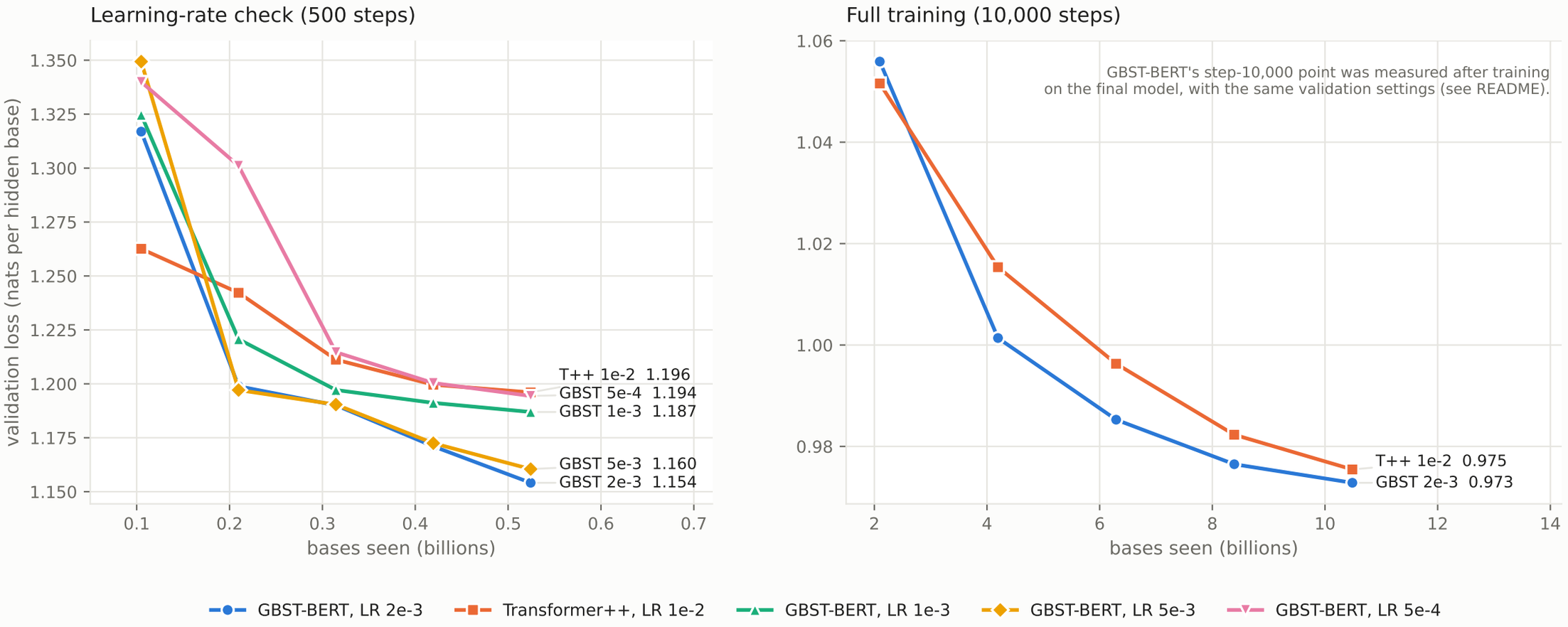
Validation loss for every trained learning rate (lower is better; random guessing = 1.386)



	Model	Learning rate	Steps trained	Val. loss at 0.5 B bases	Val. loss at 10.5 B bases	Final test loss	Final test accuracy	Note
●	GBST-BERT	2e-3	500 + 10,000	1.154	0.973	**0.9753**	**56.44%**	best in the check; used for the full run
◆	GBST-BERT	5e-3	500	1.160	–	–	–	second best
▲	GBST-BERT	1e-3	500	1.187	–	–	–	
▼	GBST-BERT	5e-4	500	1.194	–	–	–	ChaRNABERT paper value (Table 2)
■	Transformer++	1e-2	500 + 10,000	1.196	0.975	**0.9774**	**56.33%**	comparison model; Bio-xLSTM value (Table A3)
	Random guessing			1.386	1.386	1.386	25.00%	reference

How to read it: loss = how wrong the guesses of hidden bases are (lower is better); accuracy = share of hidden bases guessed correctly on unseen test DNA.
Left panel: 500-step check of four GBST-BERT learning rates (2e-3 best). Right panel: the full 10.5-billion-base runs, trained only at each model's chosen learning rate.
Result: GBST-BERT 56.44% vs Transformer++ 56.33% test accuracy — a 0.1-point difference, too small to call a winner after one run per model.